A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, IV: Local power maps, toral positivity, and conditional global comparison

Concluding Draft 6

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Abstract

We conclude the fourth part of the metaplectic Jacquet–Langlands series with a precise separation of proved local and arithmetic results from two still-conditional goals. We audit the ambient cocycle and scalar-central parameter, recall the all-torus good-element criterion, compute kernels and Haar moduli of nth-power maps on finite etale tori modulo scalars, and classify globally good regular rational classes. Building on Part III's unitary division-side character transform, we prove that wild irreducible-character integrality is equivalent to positive definiteness of every transferred irreducible character. A new toral permutation-character theorem establishes positive integral Fourier coefficients of the discriminant ratio for *every quadratic field extension* over a nonarchimedean field of odd residue characteristic, for all odd n including $p \mid n$. The complete numerical $r = 2$, $n = 3$ example and the nonabelian gluing problem are developed separately in a standalone companion paper, here begun as Draft 0. We compare our normalizations with Zou's correction of Flicker–Kazhdan, state explicit conditions for split-place matching, and prove a conditional geometric and trace-distribution comparison with normalization factor Λ. No full wild-place irreducible correspondence, construction of exceptional local matching functions, or unconditional ordinary-to-covered global comparison is claimed.

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1 Introduction and mathematical status

The problem concerns three different local representation theories. Write $H_v = D_v^\times$, and let $\overline{G_{D_v}}$ denote its specified n-fold central cover; the cover $\overline{G}_v$ of $\mathrm{GL}_r(K_v)$ is defined inside the same ambient cover. An additional *ordinary* group H_v enters through the transform $L_{D,v}$ of Part III. These are not interchangeable. In particular, a comparison between $\overline{G_D}$ and $\overline{G}$ has already been obtained in Parts I–II (here cited as Draft 8 of each) under their stated test-function and spectral hypotheses. We cite Part III as Draft 9 throughout. The new proposed comparison between H and $\overline{G_D}$ is a different assertion. This concluding version records the separate rank-two wild case in a companion manuscript [11].

Global assumptions. Let K be a totally imaginary number field with $\mu_n \subset K$, $n \geq 1$ odd. Let $r \geq 2$ be prime, $(r, n) = 1$, let L/K be cyclic of degree r, and let D be a central division K-algebra of degree r containing L. Put $S = \{v : D_v \text{ is division}\}$. Fix integers r', c such that

$$rr' \equiv 1 \pmod{n}, \qquad r - 1 + 2c \equiv 0 \pmod{n}, \qquad c_v \equiv r'c \pmod{n}. \tag{1}$$

Thus $r - 1 + 2rc_v \equiv 0 \pmod{n}$. In fact c_v is independent of v with these choices; the subscript recalls Part I's notation. Locally $F = K_v$ is a nonarchimedean field containing μ_n; p denotes its residue characteristic. We call v *wild* if $p \mid n$.

Status convention. **[P]** denotes a proof included here (possibly invoking **[E]** inputs established in cited publications or earlier parts); **[O]** denotes a proof outlined but not fully established, and **[C]** denotes a conjectural goal requiring new mathematical input. When a proved proposition has explicit unresolved hypotheses, we mark its formal implication [P] and mark the outstanding hypotheses [C] separately, in agreement with the status usage of Part III, Draft 9.

Assertion	Precise basis	Status
Diagonal and nonsplit good-element criterion, including wild p	[FK, §3]; Part I, §3	[E]
Compatibility of the Part I cocycle with σ_{c_v}	Part I, Prop. 2.3(b); direct calculation	[P]
Power-map kernel, index, and Haar modulus on $T_E(F)/F^\times$	Section 4	[P]
Global good-regular class criterion	Section 5; Part III Hasse principle	[P]
Division-side unitary character transform and tame integral case	Part III, Thms. 4.3, 7.6	[E]
Quadratic toral permutation-character theorem (all odd n)	Theorem 6.2; detailed example in companion paper	[P]
Wild irreducible-character integrality for $\widetilde{D^\times}$	Section 6	[C]
Covered D–covered GL global simple trace comparison	Parts I–II under their hypotheses	[E]
Covered D–ordinary D global comparison	Conditional implication in §8; full matching open	[P]/[C]
Removal of auxiliary support and packet hypotheses	Section 9	[C]

The running example

The series' running example is $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, which is ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$ (Part I, §1, and `notes/mjl_running_example.tex` in the project repository). Its wild place $\mathfrak{p}_3$ has $K_v = \mathbb{Q}_3(\zeta_3)$, and there D is the quaternion division algebra treated in Remark 6.3 and in the companion paper [11]. Theorem 6.2 gives the decompositions on all three quadratic tori at $\mathfrak{p}_3$. Theorem 5.2 describes the globally good classes, and Example 8.8 is the global normalization ledger for this D, in which $\Lambda = 1$ forces $\lambda_\infty = 1/3$. The smallest open case of wild integrality is whether $L_D\mathbf{1}$ at $\mathfrak{p}_3$ is irreducible of dimension 27.

2 The exact cocycle convention and star map

This section fixes the one place where a parameter mismatch could invalidate every subsequent constant. The ambient construction is a restriction of a cocycle over L_v, whereas [FK] uses the standard $\mathrm{GL}_r(F)$ Kazhdan–Patterson convention.

2.1 Restriction and relabeling

Let σ_F be the parameter-zero Kazhdan–Patterson cocycle with restriction to the diagonal torus

$$\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j} (a_i, b_j)_{n,F}. \tag{2}$$

The GL subgroup of the ambient cover in Part I has, *as an extension*, cocycle model

$$\sigma_F(g,h)^r \, (\det g, \det h)_{n,F}^{c} \quad \text{before relabeling } \mu_n. \tag{3}$$

This assertion is Part I, Proposition 2.3(b). Applying the kernel relabeling $\zeta \mapsto \zeta^{r'}$ yields the model

$$\sigma_{c_v}(g,h) = \sigma_F(g,h)(\det g, \det h)_{n,F}^{c_v}. \tag{4}$$

The two parameters c and c_v must be kept separate: c occurs before relabeling and $c_v = r'c \pmod n$ afterward. The isomorphism in (3) is an isomorphism of extensions, not a claim that arbitrary choices of cocycle sections give literally equal functions on all of $\mathrm{GL}_r(F)^2$.

Proposition 2.1 ([**P**] Diagonal commutator audit)**.** *For* $t = \mathrm{diag}(t_1, \dots, t_r)$ *and* $u = \mathrm{diag}(u_1, \dots, u_r)$ *in* $\mathrm{GL}_r(F)$*,*

$$[\widetilde{t}, \widetilde{u}] = \prod_{i,j} (t_i, u_j)_{n,F}^{A_{ij}}, \qquad A = -I_r + (1 + 2c_v)J_r. \tag{5}$$

Here J_r *is the all-ones matrix. Its row sums are* $r - 1 + 2rc_v \equiv 0 \pmod{n}$*, so scalar lifts commute with the lifted split torus. The radical on* $(F^\times/F^{\times n})^r$ *consists precisely of diagonal classes.*

Proof. The commutator is the ratio $\sigma_{c_v}(t, u)/\sigma_{c_v}(u, t)$. The Hilbert symbol satisfies $(a, b)_n = (b, a)_n^{-1}$; this is *antisymmetry*, valid for all n. The determinant twist contributes $2c_v$ to each entry of A, and the parameter-zero factor contributes 1 off the diagonal. This gives (5). The Hilbert pairing on $X = F^\times/F^{\times n}$ is perfect. As X is killed by n and $(r, n) = 1$, the decomposition $X^r = \Delta X \oplus \{(x_i) : \sum x_i = 0\}$ holds by averaging with r^{-1} modulo n. On these summands A acts, respectively, by $r - 1 + 2rc_v = 0$ and -1. Therefore $\ker A = \Delta X$. □

Remark 2.2. The assumption $(r, n) = 1$ also follows algebraically from $n \mid r - 1 + 2rc_v$, because any common divisor of r and n must divide 1. We retain it because it is a global hypothesis of the series. The oddness of n is used for the ordinary star-map convention below, not for Hilbert-symbol antisymmetry.

2.2 Star and central characters

For an element h of the ordinary group, choose any lift $\widetilde{h}$ and define $h^* = \widetilde{h}^{\,n}$. It is independent of the lift because μ_n is central. In the present scalar-central cover, for $a \in F^\times$,

$$(ah)^* = a^* h^*, \qquad a^* = \widetilde{a}^{\,n}. \tag{6}$$

If h_1, h_2 commute in an ordinary torus, their lifted commutator is in μ_n, and the class-two power identity

$$(\widetilde{h}_1\widetilde{h}_2)^n = \widetilde{h}_1^n \widetilde{h}_2^n [\widetilde{h}_2, \widetilde{h}_1]^{n(n-1)/2} = \widetilde{h}_1^n \widetilde{h}_2^n \tag{7}$$

uses n odd. Thus star is multiplicative on the lifted torus. This agrees with the convention in [FK, §4] and [Zou, §3]: Zou writes $x^* = u(x)s(x)^n$, where $u(x) = \pm 1$ is the correction defined in [2, §4] and [3, §3.1]. For odd n the extra sign is absent in the Flicker–Kazhdan convention: the star is the nth power of a lift. The correction $u(x)$ is introduced for the even-n convention; we do not infer that an arbitrary sign automatically lies in μ_n. In the odd-degree setting, therefore, $u(x) = 1$ by definition of the relevant convention (compare [2, §4] and [3, §3.1]). For a scalar $\zeta \in \mu_n(F)$, $\zeta^* = 1$ in the standard diagonal cocycle model: the self-symbol $(\zeta, \zeta)_n = 1$ for odd n, so $s(\zeta I_r)^n = s(I_r) = 1$. This last statement is also invariant under the extension isomorphisms in Part I. For representation characters we first fix a genuine unitary character $\widetilde{\theta}$ of the central preimage $\widetilde{Z}$ and *then* put $\theta(a) = \widetilde{\theta}(a^*)$. The reverse implication is not unique: θ alone does not specify $\widetilde{\theta}$. For $\zeta \in \mu_n(F)$ we have $\zeta^* = 1$, hence $\theta(\zeta) = 1$. Globally the rational splitting gives $a^* = \iota(a^n)$ for $a \in K^\times$; if $\widetilde{\theta}$ is automorphic, it is trivial on the rational splitting, so $\theta(a) = 1$ on $K^\times$. Test functions use the inverse central characters θ^{-1} and $\widetilde{\theta}^{-1}$.

3 Good elements for every maximal torus

A regular maximal torus in $\mathrm{GL}_r(F)$ may be nonsplit even at a place where $D_v \simeq M_r(F)$. This is the situation not treated by the diagonal matrix alone. The following calculation is not new: it is the proof of [FK, §3, Proposition] in the notation of this series, and also underlies Part I, Proposition 3.5.

Theorem 3.1 ([**E**] Flicker–Kazhdan good-element theorem)**.** *Let $E = \prod_{i=1}^{k} E_i$ be a finite* étale *F-algebra of dimension r, embedded as a regular maximal torus $T_E(F) = E^\times$ in $\mathrm{GL}_r(F)$. Under (1), for any $t \in T_E(F)$ and any lift $\widetilde{t}$,*

$$[\widetilde{t}, \widetilde{u}] = (\mathrm{N}_{E/F}\, t, \mathrm{N}_{E/F}\, u)_{n,F}^{1+2c_v} \prod_{i=1}^{k} (t_i, u_i)_{n,E_i}^{-1}. \tag{8}$$

The radical of this commutator pairing is

$$\{t \in T_E(F) : [\widetilde{t}, \widetilde{u}] = 1 \ \forall u \in T_E(F)\} = F^\times T_E(F)^n. \tag{9}$$

For regular t, this is exactly the criterion that $\widetilde{t}$ is good. Both identities are valid when $p \mid n$.

Proof from the cited formula, included for clarity. The commutator formula is [FK, §3], using [KP, Prop. 0.1.5]; its restriction to a single field is exactly Part I, Proposition 3.1 after relabeling [9]. Write $N = \mathrm{N}_{E/F}$. Norm compatibility of local Hilbert symbols gives

$$(Nt, Nu)_{n,F} = \prod_i (Nt, u_i)_{n,E_i}.$$

Thus (8) equals

$$\prod_i \left((Nt)^{1+2c_v} t_i^{-1}, u_i\right)_{n,E_i}.$$

The nondegeneracy of the nth Hilbert symbol over each E_i (including wildly ramified E_i) implies that the commutator is trivial for all u exactly when

$$t_i \equiv (Nt)^{1+2c_v} \pmod{E_i^{\times n}} \quad (1 \le i \le k). \tag{10}$$

This directly says $t \in F^\times E^{\times n}$. Conversely, if $t = ay^n$ for $a \in F^\times$, then $Nt = a^r (Ny)^n$; therefore $(Nt)^{1+2c_v} \equiv a^{r(1+2c_v)} \equiv a \equiv t_i$ modulo $E_i^{\times n}$ by scalar centrality. This proves the criterion. No tameness assumption has been introduced. □

Example 3.2 (The diagonal rank-two check)**.** Take $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$. In the notation of Part I, Example 3.7, choose $c = 1$, so $c_v \equiv 2 \pmod 3$. Formula (5) becomes

$$A \equiv \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \pmod 3, \qquad [\widetilde{t}, \widetilde{u}] = (t_1/t_2, u_1/u_2)_{3,F}.$$

Thus $\mathrm{diag}(t_1, t_2)$, when regular, is good exactly when $t_1/t_2 \in F^{\times 3}$. If a quadratic field E/F is embedded as a nonsplit maximal torus, the same theorem says that goodness is equivalent to $t \in F^\times E^{\times 3}$; the two tests are instances of the same formula, not separate conjectures.

Remark 3.3 (Why goodness is not enough for a regular-root formula)**.** The power of a regular element in a *split* torus need not be regular. For $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, the element $\mathrm{diag}(1, \zeta_3)$ is regular but not good: $\zeta_3 \notin F^{\times 3}$, since F does not contain ζ_9. Its cube is the identity. A character or orbital-integral formula with $\Delta(t^n)$ in the denominator must restrict to t^n regular (as in Zou's equation (3.2)), or use a separately proved germ extension. On a degree-r *field* torus, by contrast, if t is regular and $t^n \in F$, then Kummer theory gives $[F(t) : F] \mid n$; since $[F(t) : F] = r$ and $(r, n) = 1$, this is impossible. Because r is prime, $F(t^n) = F(t)$ and t^n is regular.

4 Local power-map calculation for étale tori

The key local distinction is between a field centralizer (compact modulo scalars) and a product of fields (not compact modulo scalars). Both occur in the split *group*. The next proposition gives an exact count of the roots and the size of the image at wild and tame places, including the factors that a global trace identity must account for.

Lemma 4.1 ([**P**] Kummer injection for a product algebra)**.** *Let $E = \prod_{i=1}^{k} E_i$ be finite* étale *of dimension r over F, with $\mu_n \subset F$ and $(r, n) = 1$. Then*

$$F^\times/F^{\times n} \longrightarrow E^\times/E^{\times n} \tag{11}$$

is injective.

Proof. Suppose $a \in F^\times$ is an nth power in every E_i. The Kummer extension $F(a^{1/n})/F$, since $\mu_n \subset F$, is cyclic of degree $d \mid n$ and embeds in every E_i. Therefore $d \mid [E_i : F]$ for every i, so also $d \mid \sum_i [E_i : F] = r$. Coprimality gives $d = 1$ and $a \in F^{\times n}$. □

Proposition 4.2 ([**P**] Kernel, index, and Haar modulus)**.** *Under the hypotheses of Lemma 4.1, put $A_E = E^\times/F^\times$ and $P_n([t]) = [t^n]$. Then*

$$\ker(P_n) \simeq \mu_n(E)/\mu_n(F) \simeq \mu_n^k/\mu_n, \qquad |\ker(P_n)| = n^{k-1}, \tag{12}$$

$$[A_E : P_n(A_E)] = n^{2(k-1)}|n|_F^{-(r-1)}. \tag{13}$$

In particular, $P_n(A_E)$ is open and closed. With any *Haar measure da on A_E and its restriction to $P_n(A_E)$, for compactly supported integrable q on that image,*

$$\int_{A_E} q(a^n)\, da = n^{k-1}|n|_F^{-(r-1)} \int_{P_n(A_E)} q(b)\, db. \tag{14}$$

Proof. If $t^n = a \in F^\times$, then a is an nth power in every E_i; by Lemma 4.1, write $a = b^n$ with $b \in F^\times$. Thus each $t_i/b \in \mu_n(E_i) = \mu_n(F)$, giving (12), including its canonical description.

For a local field M containing μ_n the Kummer quotient has

$$|M^\times/M^{\times n}| = n^2|n|_M^{-1}. \tag{15}$$

Indeed the discrete valuation contributes n; the finite torsion unit factor containing μ_n contributes n; and the torsion-free p-adic unit factor contributes $|n|_M^{-1}$ (multiplication by n on its $\mathbb{Z}_p$-lattice). Alternatively, (15) follows from local duality. Since $F^\times/F^{\times n}$ injects into $E^\times/E^{\times n}$, the group A_E/A_E^n is its cokernel:

$$|A_E/A_E^n| = \frac{\prod_i n^2|n|_{E_i}^{-1}}{n^2|n|_F^{-1}} = n^{2(k-1)}|n|_F^{-(r-1)},$$

where $|n|_{E_i} = |n|_F^{[E_i:F]}$ for normalized absolute values on local fields. This proves (13).

For the Haar statement, write $K_0 = \ker(P_n)$, of order n^{k-1}. Choose a sufficiently small compact open subgroup $U \subset A_E$ consisting of principal-unit classes and having $U \cap K_0 = \{1\}$. The p-adic logarithm identifies U with a free $\mathbb{Z}_p$-module of rank $(r-1)[F : \mathbb{Q}_p]$: the factors $E_i^\times$ have combined principal-unit rank $r[F : \mathbb{Q}_p]$, and quotienting by the scalar principal units subtracts $[F : \mathbb{Q}_p]$. Multiplication by n maps U injectively into itself, with

$$[U : U^n] = p^{v_p(n)(r-1)[F:\mathbb{Q}_p]} = |n|_F^{-(r-1)}.$$

Since $P_n^{-1}(U^n) = K_0U$ and $K_0 \cap U = \{1\}$, its pushforward of Haar measure gives

$$\frac{\mathrm{vol}(P_n^{-1}(U^n))}{\mathrm{vol}(U^n)} = \frac{|K_0|\,\mathrm{vol}(U)}{\mathrm{vol}(U)/[U : U^n]} = n^{k-1}|n|_F^{-(r-1)}.$$

The pushforward is translation-invariant on the open image $P_n(A_E)$, so this ratio is its constant density on that image, proving (14). Notice that this argument requires neither a global quotient-index formula nor a choice of scale for Haar measure. □

Remark 4.3 (A measure-theoretic warning)**.** For a general open endomorphism $\varphi : A \to A$ of a locally compact abelian group with finite kernel, its pushforward Haar density need not equal $[A : \varphi A]/|\ker\varphi|$. For example, on $A = \mathbb{Z}$ with counting measure, $a \mapsto na$ has density 1 on $n\mathbb{Z}$, not n. The correct local computation here is $|\ker P_n|[U : U^n]$, where U is the compact open subgroup in the proof. For this specific A_E it happens to simplify to $[A_E : P_nA_E]/|\ker P_n|$; that numerical equality is not a general change-of-variables theorem.

Example 4.4 (Three local torus types)**.** If E/F is a field of degree r, then $k = 1$ and P_n is injective, with index and Haar modulus $|n|_F^{-(r-1)}$. If $E = F^r$ then $k = r$ and the kernel has order n^{r-1}. For $r = 2, n = 3$, $F = \mathbb{Q}_3(\zeta_3)$ and $|3|_F = 1/9$,

type	$\lvert\ker P_3\rvert$	$[A_E : A_E^3]$	factor in (14)
E/F quadratic	1	9	9
$E = F \times F$	3	81	27

These integers quantify why one cannot use the division-torus substitution formula unchanged on the full split group.

5 A global classification of good regular classes

The local theorem now applies to the actual torus at *every* place. Combining it with the Hasse principle of Part III removes the local-to-global ambiguity for power classes of a degree-r field, though not the independent measure and spectral questions.

Proposition 5.1 ([**E**] Almost-everywhere powers modulo scalars)**.** *Let E/K be a field extension of degree r with $(r, n) = 1$ and $\mu_n \subset K$. For every finite set Σ of places,*

$$E^\times/(K^\times E^{\times n}) \longrightarrow \prod_{v\notin\Sigma} \frac{(E \otimes_K K_v)^\times}{K_v^\times (E \otimes_K K_v)^{\times n}} \tag{16}$$

is injective. This is Part III, Proposition 11.3 [12].

Recalled proof. Let t be locally scalar times an nth power outside Σ. Choose $u, k \in \mathbb{Z}$ with $ru+nk = 1$ and put $a = N_{E/K}(t)^u \in K^\times$. If $t = a_v y_v^n$ in the local étale algebra, taking norms gives the identity

$$t/a = \left(a_v^k y_v / N_{E_v/K_v}(y_v)^u\right)^n.$$

Thus t/a is an nth power in every completion of E over all $v \notin \Sigma$. If its class in $E^\times/E^{\times n}$ were nontrivial, the nontrivial Kummer extension $E((t/a)^{1/n})/E$ would, by Chebotarev, admit infinitely many unramified places at which t/a is not an nth power. This contradicts the hypothesis. Hence $t \in K^\times E^{\times n}$. □

Theorem 5.2 ([**P**] Global good-regular criterion)**.** *Assume the covers and rational splittings of Part I. Let G denote $D^\times$ or GL_r, and let $\gamma \in G(K)$ be regular elliptic, so $E = K[\gamma]$ is a field of degree r. The following are equivalent:*

(a) the rational lift $\iota(\gamma)$ is good at every place;

(b) it is good at every place outside some finite set Σ;

(c) $\gamma \in K^\times E^{\times n}$.

The map $[s] \mapsto [s^n]$ is a bijection from the noncentral classes in $E^\times/K^\times$ onto the globally good regular classes belonging to E. Furthermore

$$W_{s^n} = W_s, \qquad W_s = \{\sigma \in \mathrm{Aut}(E/K) : \sigma(s) \in K^\times s\}. \tag{17}$$

Proof. At each finite v, the ordinary centralizer is $T_v = (E \otimes_K K_v)^\times$. If D_v splits, apply Theorem 3.1, transported through Part I, Proposition 2.6 (the split-place isomorphism of the specified covers). If D_v is division, use Part I, Proposition 3.5(a), whose commutator is the same by Part I, Proposition 3.1. At complex places the nth-power map on T_v is surjective. Thus at every place the local goodness condition is precisely $\gamma \in K_v^\times T_v^n$. Implications (c)⇒(a)⇒(b) follow, while (b)⇒(c) is Proposition 5.1.

The power map $E^\times/K^\times \to E^\times/K^\times$ is injective: if $s^n \in K^\times$, Kummer theory shows $[K(s) : K] \mid n$, while $[K(s) : K] \mid [E : K] = r$. Hence $s \in K^\times$. Because r is prime, every noncentral $s \in E$ generates E, and the same is true of s^n. This proves the bijection. Finally $\sigma(s^n)/s^n \in K^\times$ if and only if $(\sigma(s)/s)^n \in K^\times$, which is equivalent by injectivity to $\sigma(s)/s \in K^\times$. Equation (17) follows. □

Remark 5.3 (Precise consequence for the geometric side)**.** Part I, Proposition 6.1, uses the rational regular classes which are good everywhere, with coefficient

$$c(\gamma) = \frac{\mathrm{vol}(\mathbb{A}_K^\times E^\times \backslash \mathbb{A}_E^\times; dt)}{|W_\gamma|}. \tag{18}$$

Theorem 5.2 parametrizes its noncentral summation set by global $[s^n]$ and preserves W_γ. It does *not* replace the local root sums in Zou's character formula: the global root class may be unique even when a localized split torus has n^{k_v-1} roots modulo scalars.

6 Wild local integrality: exact reductions and a toral theorem

The division-side transform of Part III is an isometric isomorphism of spaces of invariant class functions. Wild-place irreducible-character integrality does not follow from this analytic statement. We record here the exact character-theoretic criterion and a positive result on quadratic field tori. Detailed ramified and unramified computations for $F = \mathbb{Q}_3(\zeta_3)$ belong to the separate companion project *Wild-place metaplectic character transfer in degree two*, Draft 0, cited below as [11].

6.1 Normalization and the unitary character matrix

Let F be a nonarchimedean field containing μ_n, let D/F be a division algebra of prime degree r with $(r, n) = 1$, and put $H = D^\times$ and $Z = F^\times$. Write $\widetilde{H}$ for the scalar-central cover specified in Part I. For a regular element t in a maximal field $E \subset D$, put

$$\Delta(t) = |\det(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})|_F^{1/2}, \qquad R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}. \tag{19}$$

The character transform of Part III, with compatible central characters $\theta, \widetilde{\theta}$, is

$$(L_D\Phi)(zt^*) = b_D\widetilde{\theta}(z)R_E(t)\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}; \tag{20}$$

it vanishes on nongood regular lifts. Its continuously extended identity value is

$$(L_D\Theta_\rho)(1) = c_D \dim \rho, \qquad c_D = |n|_F^{-(r^2-1)/2}. \tag{21}$$

Theorem 4.3 and Corollary 5.1 of Part III prove that L_D is a *unitary onto map* between the fixed-central-character invariant L^2 spaces and admit finite expansions

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle. \tag{22}$$

We do not assume that $a_{\rho\tau}$ is integral at wild places.

Proposition 6.1 ([**P**] Integrality–positivity criterion)**.** *For the entire unitary matrix $A = (a_{\rho\tau})$, the following conditions are equivalent:*

(i) L_D *bijects the irreducible-character bases with positive signs;*

(ii) $a_{\rho\tau} \in \mathbb{Z}$ *for all* ρ, τ*;*

(iii) $a_{\rho\tau} \geq 0$ *is real for all* ρ, τ*;*

(iv) every $L_D\Theta_\rho$ *is a continuous positive-definite function on* $\widetilde{H}$*.*

Positive definiteness for one *row is not sufficient for integrality of that row.*

Proof. A row with integral entries and squared norm one has exactly one nonzero entry, equal to ± 1. Equation (21) rules out the minus sign, and unitarity and surjectivity give the bijection. Conversely, permutation of the character bases gives integrality. For nonnegative entries, orthogonality of distinct rows implies that their supports are disjoint. Every column has squared norm one; it meets at most one row, and its sole nonzero entry therefore equals one. Each row has norm one, so it also contains exactly one entry equal to one. Finally, Peter–Weyl orthogonality for the compact-mod-center group with fixed unitary central character identifies positive-definite invariant functions with those having nonnegative coefficients in the irreducible-character basis. A single nonnegative norm-one row may nevertheless contain several nonintegral entries (for example $1/\sqrt{2}, 1/\sqrt{2}$). □

6.2 The toral Fourier reduction

For a maximal field E/F, $A_E = E^\times/F^\times$ is compact abelian. The continuous discriminant ratio is locally constant and hence has a finite Fourier expansion

$$R_E(t) = \sum_{\chi \in \widehat{A_E}} m_\chi \chi(t). \tag{23}$$

A genuine central character on $\widetilde{E^\times}$ compatible with an ordinary character α identifies the coefficient m_χ with the multiplicity of the corresponding genuine Stone–von Neumann torus representation in the *restricted* transfer $\mathcal{L}_E\alpha$ (Part I, Proposition 3.5, and Part III). This is a statement about the full preimage of the field torus; it is not an assertion that the torus modules extend to $\widetilde{H}$. Consequently,

$$m_\chi \in \mathbb{Z}_{\geq 0} \quad \text{for all } \chi \tag{24}$$

is sufficient for positive integral *torus restrictions* of all transferred ordinary characters. It does not imply any $a_{\rho\tau} \in \mathbb{Z}$ for the full nonabelian cover.

Theorem 6.2 ([**P**] Quadratic toral permutation positivity)**.** *Let $F/\mathbb{Q}_p$ be finite with p odd, let n be odd with $\mu_n \subset F$, and let E/F be a quadratic field extension. Then R_E is a permutation character of a finite quotient of $E^\times/F^\times$. In particular the Fourier coefficients in (23) belong to $\mathbb{Z}_{\geq 0}$, even when $p \mid n$. At the identity, $R_E(1) = |n|_F^{-1}$.*

Proof. Write $n = p^a m$ with $(m,p) = 1$ and set $G = E^1 = \ker N_{E/F}$. Hilbert 90 identifies $E^\times/F^\times$ with G by $t \mapsto u = t/\sigma(t)$. If $a = 0$ the ratio is identically one, so suppose $a > 0$. Set $G_k = \{u \in G : v_E(1-u) \geq k\}$ and $G_0 = G$. A root $\zeta \in \mu_n$ with nontrivial prime-to-p component cannot have $\zeta u \equiv 1$ modulo the maximal ideal: reduction of u has norm one and its intersection with the scalars is $\{\pm 1\}$. For $\zeta \in \mu_{p^a} \setminus \{1\}$, the ultrametric inequality, together with $\sigma(u) = u^{-1}$ and p odd, gives

$$v_E(1-\zeta u) = \min\big(v_E(1-u), v_E(1-\zeta)\big).$$

Indeed, cancellation at equal valuations would give both $u \equiv \zeta^{-1}$ and $u^{-1} \equiv \zeta^{-1}$ beyond the valuation of $1-\zeta$, contradicting $v_E(\zeta - \zeta^{-1}) = v_E(1-\zeta)$. Factoring $1-u^n$ yields

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(v_E(1-u), v_E(1-\zeta))}, \tag{25}$$

where q_E is the size of the residue field of E. This formula also defines the continuous central value.

Put $t_\zeta = v_E(1-\zeta)$, $T = \max_{\zeta \neq 1} t_\zeta$ and $M_k = q_E^{\frac{1}{2}\sum_\zeta \min(k, t_\zeta)}$ ($M_0 = 1$). The M_k are integers, since ζ and ζ^{-1} occur in pairs, and $M_k = M_T$ for $k \geq T$. If E/F is unramified and $q = q_F$, then $q_E = q^2$ and

$$[G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Writing $s_k = \#\{\zeta \neq 1 : t_\zeta \geq k\}$, we have $M_k/M_{k-1} = q^{s_k}$ and s_k is even. The quantities

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T)$$

are nonnegative integers: $q+1 \mid q^{s_k} - 1$, and when $s_k > 0$, the preceding M_{k-1} is divisible by q^{k-1}. Therefore

$$R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k [G : G_k] \mathbf{1}_{G_k}.$$

Each summand $[G : G_k]\mathbf{1}_{G_k}$ is a permutation character of G/G_k.

If E/F is ramified, then $q_E = q$, the t_ζ are even, and $G_{2j} = G_{2j+1}$, with $[G : G_1] = 2$ and $[G : G_{2j}] = 2q^j$ for $j \geq 1$. Set

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j}.$$

The first is a nonnegative integer because M_1 is odd. When the numerator of c_{2j} is nonzero, some pair ζ, ζ^{-1} has $t_\zeta \geq 2j$; hence $q^j \mid M_{2j-1}$. The quotient M_{2j+1}/M_{2j-1} is a power of odd q, so its difference from one is even. Thus every $c_{2j} \in \mathbb{Z}_{\geq 0}$ and

$$R_E = \mathbf{1}_G + c_1[G : G_1]\mathbf{1}_{G_1} + \sum_{1 \leq j \leq T/2} c_{2j}[G : G_{2j}]\mathbf{1}_{G_{2j}}.$$

The norm-one filtration indices follow directly by writing $E = F(\sqrt{d})$, $v_F(d) = 1$, and parameterizing G_1 by $(1+c\sqrt{d})/(1-c\sqrt{d})$ with $c \in \mathcal{O}_F$; congruence modulo G_{2j} records c modulo $\mathfrak{p}_F^j$. This completes the two cases. At the center $R_E(1) = |n|_F^{-1}$, by the cyclotomic product. □

Remark 6.3 ([**P**] A quantified rank-two instance)**.** For $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$ and the quaternion D/F, every quadratic maximal subfield E/F satisfies Theorem 6.2. The detailed decompositions are

$$R_{E_{\mathrm{ur}}} = 1 + 2\,\mathrm{Reg}(C_4), \qquad R_{E_{\mathrm{ram}}} = 1 + \mathrm{Reg}(C_2) + \mathrm{Reg}(C_6).$$

These give genuine torus restriction multiplicities $(3, 2, 2, 2)$ and $(3, 2, 1, 1, 1, 1)$, respectively, of degree-three constituents, hence total dimension 27. Both ramified quadratic isomorphism classes yield the same formula. The full derivations and the status of the unproved nonabelian gluing step are recorded in [11].

6.3 What the local reductions prove, and what they do not

Theorem 6.2 establishes toral Fourier positivity for *all* quadratic extensions and odd covering degrees, including wild primes, but does not establish positive definiteness on the full cover. The logical implications are

$$\text{positive character on every covered torus} \quad \not\Longrightarrow \quad \text{positive-definite character on } \widetilde{H}.$$

At a fixed finite quotient this is visible even for $C_2 \times C_2$: the function equal to 2 at the identity and -2 at each of the three involutions restricts to twice the sign character on every cyclic subgroup of order two, but equals $-\chi_0 + \chi_1 + \chi_2 + \chi_3$ on the full group, so it is not positive definite. (The function equal to 2 at the identity and 0 elsewhere shows separately that positivity does not force integrality: it restricts to the regular character on each cyclic subgroup, but is half the sum of the four irreducible characters.) The finite-level Brauer/Clifford and induced-type strategies in [11] address this additional *nonabelian gluing* requirement. For arbitrary prime $r > 2$, even the toral Fourier-positivity step remains open. No theorem here identifies a wild irreducible genuine image or proves the new ordinary-$D^\times$ to covered-$D^\times$ automorphic comparison unconditionally.

7 Connection with Zou's corrected character relation

For the Kazhdan–Patterson cover of the *split group*, Zou [3, §3] revisits the Flicker–Kazhdan metaplectic correspondence and records a corrected scalar in the character relation. This is an essential input, but its objects are $\widetilde{\mathrm{GL}_r(F)}$ and ordinary $\mathrm{GL}_r(F)$, not directly $\widetilde{H}$ and H at a division place.

Proposition 7.1 ([**E**] Zou's root-sum constant in our case)**.** *Write $d_F(m) = m|m|_F^{-1/2}$ and $d_r = \gcd(n, 2rc_v + r - 1)$. Under (1), $d_r = n$. In Zou's equation (3.2), the coefficient preceding the root sum in rank r is*

$$\frac{1}{d_F(d_r)|n|_F^{r/2}} = \frac{1}{n|n|_F^{(r-1)/2}} = \frac{b_D}{n}. \tag{26}$$

For regular elliptic x with degree-r field centralizer, x^n is automatically regular by Remark 3.3. The equation $t^n = x^n$ then has exactly n solutions $t = \zeta x$, $\zeta \in \mu_n$. If the linear character π has central character ω_π trivial on μ_n, these summands agree and the effective coefficient is b_D. This restriction on ω_π is essential: without it, the scalar sum is $\sum_{\zeta \in \mu_n} \omega_\pi(\zeta) = 0$, so equation (3.2) instead forces the covered character to vanish at x^*. For a genuine lift pair the compatibility $\omega_\pi(a) = \omega_{\widetilde{\pi}}(a^*)$ and $\zeta^* = 1$ imply $\omega_\pi|_{\mu_n} = 1$ automatically.

Proof. The equation $d_r = n$ is immediate from scalar centrality; (26) follows by substitution. Regularity of x^n follows from the field case of Remark 3.3. Any solution t centralizes x^n, so

$t \in F[x]^\times$. In the field $F[x]$, $(t/x)^n = 1$, and $\mu_n(F[x]) = \mu_n(F)$. Thus $t = \zeta x$. The Weyl discriminant is invariant under central scalar multiplication, the ordinary character takes the same value because it is trivial on μ_n, and $(\zeta x)^* = x^*$ by the explicit odd-n scalar star computation in Section 2.2. These are precisely the equal terms in Zou's (3.2). □

Remark 7.2 (What is and is not composed)**.** At the ramified places one may compare the *desired* $\overline{G_{D,v}} \leftrightarrow \overline{G}_v$ character relation of Parts I–II with (a) the classical linear Jacquet–Langlands elliptic character identity and (b) Zou's formula. The common coefficient b_D and the familiar sign $(-1)^{r-1}$ are consistent with this comparison. Nevertheless *the existence of an irreducible genuine image on the covered division side is the missing conclusion*; combining two known formulas cannot be used to assume that missing representation exists. For non-elliptic split centralizers, retain Zou's entire root sum, including the central-character factors.

8 Global geometric comparison: proved input and new obligations

There are two comparisons to keep distinct:

$$\begin{aligned} \overline{G_D}(\mathbb{A}_K) &\longleftrightarrow \overline{G}(\mathbb{A}_K) \quad \text{(Parts I–II, under their hypotheses)}, \\ D^\times(\mathbb{A}_K) &\dashrightarrow \overline{G_D}(\mathbb{A}_K) \quad \text{(proposed additional comparison)}. \end{aligned} \tag{27}$$

The first has a proved simple trace-formula comparison with regular-support or later matching hypotheses (Part I, Theorem 6.6; Part II, Theorem 6.2). The second is not a formal consequence of the first or of the unitarity of L_D.

8.1 Geometric coefficients and the product formula

For a regular elliptic rational γ, with $E = K[\gamma]$, Part I identifies the torus coefficient $c(\gamma)$, recalled in (18). The same torus measure dt must be used whenever this term is compared with the corresponding linear inner-form orbital integral. Part I uses the normalized orbital integral $J_v(\gamma, f) = \Delta_v(\gamma) O_v(\gamma, f)$. For a rational regular element the product formula implies

$$\prod_v \Delta_v(\gamma) = 1. \tag{28}$$

This is because each discriminant is the normalized absolute value of a nonzero element of K (or its square root), including the determinant normalization in (19). The local *ratio* of discriminants $\Delta_v(t)/\Delta_v(t^n)$ must be kept until local transfer and the global product are both defined.

At $v \in S$ Part III, Theorem 9.1, gives the exact local normalized orbital-integral identity

$$J_{\widetilde{H}_v}(z\, t^*, \mathcal{O}_{D,v} f_v) = b_{D,v}\, \widetilde{\theta}_v(z)^{-1} J_{H_v}(t, f_v), \tag{29}$$

using the same quotient measures on the two copies of H_v/Z_v. At $v \notin S$, $H_v \cong \mathrm{GL}_r(K_v)$ and the appropriate root-sum comparison is not (29); it must use Flicker–Kazhdan–Zou, and it can involve the kernel $\mu_n^{k_v}/\mu_n$ of Proposition 4.2. In particular the local factors $b_{D,v}$ for $v \in S$ *cannot* simply be cancelled using $\prod_v |n|_v = 1$ without the split-side contributions.

8.2 Haar ledger

For $v \in S$, Part II, Definition 2.1, fixes

$$\mathrm{vol}(D_v^\times / F_v^\times; dg_{D,v}) = d(\mathrm{St}; dg_{\mathrm{GL},v})^{-1}, \tag{30}$$

where $d(\mathrm{St}; dg_{\mathrm{GL},v})$ is the formal degree relative to the chosen $\mathrm{GL}_r(F_v)/F_v^\times$ measure. The probability measure used in Part III is $d(\mathrm{St}; dg_{\mathrm{GL},v})\, dg_{D,v}$. When *both* sides of (29) are scaled by the same measure, $b_{D,v}$ is unchanged. Different measures on the GL side must still be reconciled with (30).

Item	Required statement or calculation	Status
Ambient cocycle and common center	Part I, Lemma 5.1 and Prop. 5.5; restriction (3)	[E]
Regular rational good locus	Theorem 5.2	[P]
Global rational/stable classes	Hilbert 90 and Kottwitz's framework; local embeddability into D	[E]
Global torus coefficient	Same dt in (18), including $\lvert W_\gamma\rvert$	[E]
Local $v \in S$ orbital transfer	Part III, Theorem 9.1, equation (29)	[E]
Split-place $v \notin S$ invariance	Lemma 8.2 (necessary); rank-two spherical calculation below	[P]
Split-place pointwise transfer and λ_v	Proposition 8.5; unproved existence	[C]
Unramified unit elements	[2, §§11–12] under $(n, N_r) = 1$; transport to Part I cover pending	[E]/[O]
Nonregular powers and singular germs	Remark 3.3; germ compatibility or support needed	[C]
Central identity class	Suppressed in Prop. 8.5; separate germ/measure test below	[E]/[C]
Spectral expansion	Part I–II packet formula with restrictions; new factor matching	[E]/[C]

Here N_r is the least common multiple of the *composite* positive integers at most r (the convention in [FK]), with $N_r = 1$ when there are none. The classical unramified unit matching in [2, §12, Corollary] is established under $(n, N_r) = 1$ and $p \nmid n$. Thus $N_2 = N_3 = 1$, and for prime $r = 5$ we have $N_5 = 4$, automatically prime to the odd integer n. Other ranks must retain the published hypothesis.

8.3 Split-place invariance and a geometric comparison criterion

At a place $v \notin S$ the algebra D_v is split, but a rational elliptic field $K[s]$ may decompose locally as a product of fields. By Proposition 4.2, its local nth-root fibre modulo scalars can contain n^{k_v-1} points. The global class $[s]$ supplied by Theorem 5.2 is nevertheless unique. Therefore a local sum over *all* roots cannot be substituted for the orbital integral of this one rational representative without an additional compatibility statement.

Definition 8.1 (FK-compatible ordinary functions)**.** Fix the local central characters $\theta_v, \widetilde{\theta}_v$ with $\theta_v(a) = \widetilde{\theta}_v(a^*)$. A normalized ordinary test function f_v is called *FK-compatible on the regular-power locus* if, whenever $x, x' \in H_v$ have x^n, x'^n regular and their star images satisfy $(x')^* = zx^*$ for a central lift $z \in \widetilde{Z}_v$ (after conjugating representatives), one has

$$J_{H_v}(x', f_v) = \widetilde{\theta}_v(z)^{-1} J_{H_v}(x, f_v). \tag{31}$$

The case $x' = \xi x$ with *scalar* $\xi \in \mu_n(F)$ is automatic, since $\theta_v(\xi) = 1$. The substantive condition occurs on a product torus $E_v = \prod_{i=1}^{k_v} E_i$ with $k_v \geq 2$: if $\zeta = (\zeta_i) \in \mu_n(E_v) \simeq \mu_n(F)^{k_v}$ is nonscalar and x^n is regular, then

$$J_{H_v}(\zeta x, f_v) = J_{H_v}(x, f_v). \tag{32}$$

The odd-n star is multiplicative on each lifted torus, and $\zeta^* = 1$, so $(\zeta x)^* = x^*$. The kernel on the quotient torus is $\mu_n(F)^{k_v}/\mu_n(F)$, of size n^{k_v-1} (Proposition 4.2). Equation (31) is the invariant-function condition of [2, §6], expressed with the present inverse central transformation convention.

Lemma 8.2 ([**P**] Necessary split-place invariance)**.** *Suppose a genuine $\widetilde{f}_v$ and a fixed nonzero constant λ_v satisfy*

$$J_{\widetilde{H}_v}(x^*, \widetilde{f}_v) = \lambda_v J_{H_v}(x, f_v) \tag{33}$$

for all x in a conjugation- and scalar-stable region where x^n is regular, with the identity at central translates understood with the prescribed inverse-central-character law. Then f_v obeys (31) on that region.

Proof. If $(x')^* = zx^*$, conjugation invariance and the inverse central transformation law of the genuine test function imply $J_{\widetilde{H}_v}((x')^*, \widetilde{f}_v) = \widetilde{\theta}_v(z)^{-1} J_{\widetilde{H}_v}(x^*, \widetilde{f}_v)$. Apply (33) twice and divide by the fixed nonzero λ_v. In particular, for $x' = \zeta x$ with nonscalar $\zeta \in \mu_n(E_v)$, one has $(x')^* = x^*$ and obtains (32). The scalar-root case is already automatic from the central-character convention. Thus FK-compatibility is a *necessary* condition, not an asserted construction of $\widetilde{f}_v$. □

Proposition 8.3 ([**P**] Rank-two tame spherical-unit check)**.** *Let F be nonarchimedean with $\mu_n \subset F$, $p \nmid n$, $r = 2$, and $K_0 = \mathrm{GL}_2(\mathcal{O}_F)$. Take a spherical ordinary unit f^0, supported on $F^\times K_0$, with an unramified central character, and the standard quotient measures. Then f^0 obeys (32) for all regular split-torus x with x^n regular. For a quadratic field torus $E^\times$, the root-fibre condition modulo scalars is vacuous. Moreover, when f^0 and the genuine spherical unit $\widetilde{f}^0$ are transported from the standard Flicker–Kazhdan cocycle and measure conventions, [2, §12, Corollary] establishes their regular-power matching with $\lambda_v = 1$ in that normalization. Whether the same equality holds* literally *for the selected ambient cover and global measures of Parts I–IV requires their explicit splitting and Haar identifications.*

Proof. A split regular element is conjugate to $\mathrm{diag}(a, b)$; multiplying by a scalar, first consider $a, b \in \mathcal{O}_F^\times$. The Iwasawa and unipotent-orbit calculation, with $\mathrm{vol}(\mathcal{O}_F) = 1$, yields

$$O_{\mathrm{GL}_2}(\mathrm{diag}(a, b), 1_{K_0}) = |a - b|_F^{-1}, \qquad \Delta(\mathrm{diag}(a, b)) = |a - b|_F,$$

so $J = 1$. Multiplication of either eigenvalue by a root of unity preserves its valuation, hence the normalized orbital integral. If the two eigenvalue valuations differ, neither x nor such a translate is conjugate into $F^\times K_0$; both orbital integrals vanish. Scalar translations contribute only the specified unramified central-character factor. This proves the ordinary split-torus invariance. For a quadratic field E/F, Proposition 4.2 gives $\ker(P_n : E^\times/F^\times \to E^\times/F^\times) = 1$. The final statement invokes the *published* unit matching, not the elementary split-torus calculation. Its passage to the specified ambient cover is a distinct normalization obligation. □

Remark 8.4 (Scope of the published fundamental lemma)**.** The Flicker–Kazhdan result is a theorem for the matching unramified Hecke units, not an existence result for arbitrarily prescribed functions at finitely many exceptional places. Its original hypotheses include $p \nmid n$ and $(n, N_r) = 1$. In rank two the only maximal-torus types are split and quadratic field tori, so the ordinary invariance calculation above is exhaustive. At higher ranks, mixed tori do not automatically invalidate spherical matching: the Flicker–Kazhdan theorem covers them as well whenever its hypotheses apply.

A regularity and germ obligation. Equation (33) is imposed only where x^n is regular. On a split torus, x can be regular while x^n is singular (Remark 3.3); the power map is a local analytic isomorphism on the torus, but normalized orbital integrals of a function at images approaching a singular element may have nonconstant Shalika-germ behavior. A full transfer construction must either arrange support away from these root-collision loci or prove the matching germ conditions there. The same issue occurs at the complex places, although the cover there splits: the ordinary power map can still carry regular elements to singular ones. No universal vanishing of the germ coefficients is claimed here.

Proposition 8.5 ([**P**] Conditional geometric comparison with an explicit normalization factor)**.** *Fix the rational splittings and compatible central characters of Parts I–III, and fix* corresponding *local Haar measures, global torus measures, and normalizer conventions on the ordinary and covered sides. Choose factorizable ordinary and genuine test functions in a trace-class, regular elliptic support range for which the two simple trace formulas apply. Assume:*

(i) For each $v \notin S$, f_v is FK-compatible as in Definition 8.1, and a genuine test function $\widetilde{f}_v$ satisfies (33) for every regular rational local component s_v of a globally good class $[s^n]$, with a nonzero constant λ_v independent of s_v and of its local torus type. The prescribed central-translate identity holds, $\lambda_v = 1$ for almost all v, and the hypotheses include the complex places.

(ii) For each $v \in S$, use the Part III, Theorem 9.1, orbital-integral transfer $\widetilde{f}_v = \mathcal{O}_{D,v} f_v$, with coefficient $b_{D,v} = |n|_v^{-(r-1)/2}$ and matching quotient measures on H_v/Z_v and $\widetilde{H}_v/\widetilde{Z}_v$. The functions obey the reciprocal central transformation laws.

(iii) The common rational splitting and all local lifts satisfy the star compatibility $\iota(s^n) = \iota(s)^n$; the global torus coefficients (including $|W_s|$), central phases, and quotient measures agree under the stipulated identifications. The support condition removes the central identity contribution. The ordinary and covered simple geometric expansions are absolutely convergent in the selected test-function range.

Put

$$\boxed{\Lambda = \Big(\prod_{v \notin S} \lambda_v\Big)\Big(\prod_{v \in S} b_{D,v}\Big).} \tag{34}$$

Then the regular elliptic geometric distributions, and hence the corresponding simple spectral trace distributions, satisfy

$$\boxed{I_{\widetilde{H}}(\widetilde{f}) = \Lambda\, I_H(f).} \tag{35}$$

In particular the distributions agree if $\Lambda = 1$.

Proof. Since the rational splitting ι is a homomorphism, $\iota(s^n) = \iota(s)^n$; at every completion, $\iota(s)_v$ is a lift of s_v, so this equals its star lift s_v^*. By Theorem 5.2, globally good noncentral rational classes are parametrized by the unique global root class $[s]$ of $[s^n]$, preserving normalizer orders. Every ordinary regular rational class $[s]$ is paired with its good covered star class $[s^n]$; there is no additional hypothesis that the ordinary class $[s]$ itself be good. The cover has zero regular orbital integrals at nongood lifts, which contribute nothing to the covered geometric expansion. Local compatibility handles the multiple *local* roots without introducing additional rational summands. Hypothesis (i) supplies $\lambda_v J_{H_v}(s_v, f_v)$ off S, and (ii) supplies $b_{D,v} J_{H_v}(s_v, f_v)$ on S. By (iii) the remaining torus coefficients and central phases agree, and the product of Weyl discriminants is 1 by (28). Every

regular rational term on the covered side is thus Λ times the corresponding ordinary term. The central identity has been suppressed, and the stipulated trace-class and convergence conditions permit summation and the two trace formulas. This gives (35). The proof supplies *no* existence theorem for the matching functions or their exceptional local constants. It only establishes the implication under the explicit hypotheses. □

Remark 8.6 (Normalization versus canonical matching)**.** The condition $\Lambda = 1$ is a normalization once nonzero matching functions are known: rescaling one $\widetilde{f}_v$ rescales its λ_v and hence Λ. It is *not* a theorem about canonically normalized units; rescaling a unit changes its canonical value. The meaningful unresolved question is to compute the constants for the prescribed local transfers, especially the finitely many exceptional split places, and to verify the global product in those fixed conventions. Likewise, compactness of $D^\times(K)Z(\mathbb{A}_K)\backslash D^\times(\mathbb{A}_K)$ eliminates continuous spectrum but does not by itself justify every unrestricted trace sum; the support and trace-class conditions above are retained for this reason.

8.4 The identity class and an explicit normalization test

Because r is prime and D is division, every noncentral rational element generates a degree-r field and is regular elliptic. Thus, after quotienting by the rational center, the central identity class is the only remaining singular rational class. Proposition 8.5 suppresses it, but it can be used as an additional test *if* the test range is enlarged and the necessary local germ identities are supplied.

Remark 8.7 (Central discriminants versus orbital-integral germs)**.** Let $d = r(r-1)/2$. For any torus, as a regular $x \to 1$ with x^n regular, the *discriminant ratio* satisfies

$$\frac{\Delta(x)}{\Delta(x^n)} \longrightarrow |n|_v^{-d},$$

because $(1-u^n)/(1-u) \to n$ for each root value $u \to 1$. However this identity *does not by itself yield* $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ from (33). On the split group, normalized orbital integrals generally have nontrivial Shalika germs near singular central elements; their limiting coefficients need not be the evaluation of the test function at 1. The requisite relation between *central test-function values* is therefore a separate local matching obligation, even when the regular orbital integrals match. Compare the unit-specific calculation of Proposition 8.3 with the general germ warning following Lemma 8.2.

At $v \in S$, Part III, Corollary 9.4 gives $\widetilde{f}_v(1) = c_{D,v} f_v(1)$ with $c_{D,v} = b_{D,v} |n|_v^{-r(r-1)/2}$. If, as an *additional unproved local identity-value hypothesis*, every split place (including complex places) satisfies $\widetilde{f}_v(1) = \lambda_v |n|_v^{-d} f_v(1)$ with $d = r(r-1)/2$, and all denominators are nonzero, the ratio of global identity-function values becomes

$$\prod_v \frac{\widetilde{f}_v(1)}{f_v(1)} = \Lambda \prod_v |n|_v^{-r(r-1)/2} = \Lambda \tag{36}$$

by the product formula. This does not imply equality of the *identity trace-formula terms* without a separate choice of matching quotient measures: homeomorphic adelic quotients need not have equal volumes with arbitrarily scaled Haar measures. Nor does it justify retaining the identity term in the simple trace formula without checking its other hypotheses.

Example 8.8 (A possible global numerical ledger)**.** Let $K = \mathbb{Q}(\zeta_3)$, $r = 2$, $n = 3$, and choose a quaternion division K-algebra ramified at the place over 3 and at one other finite place; for instance

the running example $D = (-1, 4 - \zeta_3)_K$, ramified at $\mathfrak{p}_3$ and $\mathfrak{p}_7$. At the wild division place, $b_{D,v} = 3$, $c_{D,v} = 27$. If the finite split places have matching $\lambda_v = 1$, the equation $\Lambda = 1$ would require $\lambda_\infty = 1/3$ at the unique complex place. Since the normalized complex absolute value has $|3|_\infty = 9$, the hypothesized central-value formula would then give the complex identity factor $(1/3)/9 = 1/27$, balancing 27. This is a *necessary numerical consistency check* under the stated assumptions, not a proof of matching complex test functions or a global comparison.

8.5 Why Li, Kottwitz, and DKV do not close the gap

The linear Jacquet–Langlands theorem of Deligne–Kazhdan–Vignéras [1] controls the ordinary $D^\times$–GL_r relation, including the elliptic character sign, but it does not automatically manufacture a genuine representation of $\overline{G_{D_v}}$. Kottwitz's analysis of stable conjugacy and relative trace coefficients [5] provides the correct arithmetic setting; for the induced tori $\mathrm{Res}_{E/K}\,\mathbb{G}_m$ one has $H^1(K, T) = 1$ by Shapiro's lemma and Hilbert 90, and embeddings $E \hookrightarrow D$ are controlled by Brauer splitting. Wen–Wei Li proves an invariant trace formula for adelic covers under an invariant trace Paley–Wiener hypothesis for Levi subgroups at *real* archimedean places [7]. Here K is totally imaginary, so that real-place hypothesis is vacuous. At the complex places the specified covers split (Part I, Appendix B, item (R4); see also its ambient-cover construction), and the linear invariant Paley–Wiener theorem of Clozel–Delorme [6] supplies the relevant archimedean background. Li also discusses known real metaplectic GL cases, but his general result does not assert the particular *equality* of ordinary-$D^\times$ and covered-$D^\times$ traces required in (27). The simpler already-proved trace formulas of Parts I–II remain the appropriate initial comparison.

9 Spectral separation and a research program

9.1 The actual spectral distribution identity

Let $H = D^\times$ and $\widetilde{H} = \overline{G_D}$ globally, with compatible central characters $\theta, \widetilde{\theta}$. The quotient $H(K)Z_H(\mathbb{A}_K)\backslash H(\mathbb{A}_K)$ is compact because D is a division algebra over K; its covered analogue is compact modulo the specified central preimage as well. Thus the fixed-central-character right regular representations have discrete Hilbert decompositions. For factorizable test functions with the inverse central transformation law, define

$$I_H(f) = \sum_{\rho \in \mathcal{A}(H,\theta)} m_H(\rho) \prod_v \operatorname{tr} \rho_v(f_v), \tag{37}$$

$$I_{\widetilde{H}}(\widetilde{f}) = \sum_{\widetilde{\rho} \in \mathcal{A}(\widetilde{H},\widetilde{\theta})} m_{\widetilde{H}}(\widetilde{\rho}) \prod_v \operatorname{tr} \widetilde{\rho}_v(\widetilde{f}_v). \tag{38}$$

Here $\mathcal{A}$ denotes the relevant automorphic spectrum and m is its actual spectral multiplicity, not an assumed 1. The expressions are interpreted on trace-class test functions satisfying the simple trace-formula hypotheses; a bare unrestricted infinite sum is not justified by notation alone. Under the assumptions of Proposition 8.5, the comparison of geometric expansions supplies $I_{\widetilde{H}}(\widetilde{f}) = \Lambda I_H(f)$; ordinary equality holds when the fixed normalizations give $\Lambda = 1$. It does not identify summands. This is a conditional identity of *distributions*, with both the multiplicities and local trace factors retained.

9.2 Individual local characters and integrality

An equality of global spectral sums is a statement about *distributions*. To isolate a single local irreducible at a wild place, one needs an argument that the relevant coefficients are integers (or

multiplicities which can be shown to isolate a single constituent). Linear independence alone can separate finite lists of characters; it does not supply the required integrality from a unitary Fourier transform.

Remark 9.1 (What an integer spectral coefficient would prove)**.** If a justified global-to-local argument supplies $a_{\rho\tau} \in \mathbb{Z}$ for every τ in the row of a fixed ρ, Proposition 6.1 (Part III, Corollary 5.2) forces $L_{D,v}\Theta_\rho$ to be one irreducible genuine character with positive sign. If integrality holds for all rows, unitarity implies a bijection. This is an application of an already proved lattice criterion, not a second conditional theorem; the missing step is precisely the integrality hypothesis at a wild place.

Separating the remaining tasks. First, Theorem 6.2 proves toral permutation positivity in rank two for every odd covering degree. The separate rank-two project [11] addresses the remaining nonabelian gluing problem and must justify the choice of a finite level before any finite character calculation becomes decisive. Second, at split places the root sum in Zou's (3.2) should be written against the kernel and index in Proposition 4.2, including the chosen Haar measures and central phases. Third, one needs the explicit ambient-cover identification that transports the known unramified unit matching of [2, §12] to the selected convention, together with functions at the finitely many exceptional split places and a complete global coefficient identity as in Proposition 8.5. Every ordinary rational regular class is paired with the star of its global representative; the ordinary class itself need not be good. Finally, one must justify globalization and spectral separation for representations of the covered inner form, and remove the auxiliary supercuspidal and non-cancellation hypotheses of Parts I–II. These are mathematical work packages, not restatements of an already-proved theorem.

10 Selected glossary

Good element A regular lift commuting with the full preimage of its ordinary centralizer; genuine invariant characters vanish on the nongood regular locus.

Scalar-central cover
A cover whose entire inverse image of $F^\times I_r$ is central; here forced by $n \mid r-1+2rc_v$.

Power-map kernel
$\ker(P_n)$ on $T(F)/F^\times$, distinct from the image index. The kernel can be nontrivial on product tori even when the corresponding field-torus map is injective.

Star map $h^* = \widetilde{h}^{\,n}$, independent of lift; multiplication on tori uses n odd.

Weyl discriminant
The factor $\Delta = |\det(1 - \mathrm{Ad}(t)|\mathfrak{g}/\mathfrak{t})|^{1/2}$; it is not the unnormalized characteristic-polynomial discriminant.

Toral Fourier test
Expand the discriminant ratio R_E in characters of the compact abelian group $E^\times/F^\times$. Integral positive coefficients ensure torus-restriction positivity, not full-group integrality.

Nonabelian gluing
The additional requirement that compatible genuine covered-torus characters arise from an integral character of the full group $\widetilde{D^\times}$.

Twisted finite quotient
A finite quotient of H/Z with the central character retained as a multiplier, used to compute projective/genuine irreducible characters.

Packet identity
An identity of traces over collections of automorphic representations; it does not assert a bijection between individual local irreducibles.

Formal degree The Plancherel normalization of a square-integrable representation relative to a specified quotient Haar measure.

Root-sum relation
Zou's equation (3.2), which sums over all nth roots satisfying the stated regularity condition rather than choosing one arbitrarily.

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